Advanced Digital Signal Processing Architecture for an Ultra-Low-Latency 8-Bit Chiptune Synthesizer and Modular Effects Chain

Introduction to the Architectural Paradigm

The development of a native, ultra-low-latency 8-bit chiptune synthesizer paired with a programmable, step-automated multi-effects processor represents a rigorous intersection of modern digital signal processing (DSP) and retro-aesthetic sound design. The primary objective is to authentically recreate the sonic imperfections, limited-resolution character, and specific circuit behaviors of classic hardware—such as the Nintendo Entertainment System (NES) APU, the Commodore 64 SID chip, and the Yamaha OPL series—without inheriting the severe digital aliasing artifacts that plague naive implementations.

Simultaneously, the architecture must operate under rigid computational constraints. To remain viable as an insert instrument within modern digital audio workstations (DAWs) or embedded hardware trackers, the entire signal chain—comprising multi-oscillator polyphony, zero-delay feedback state-variable filtering, fractional delay lines, and algorithmic reverberation—must consume no more than 1 to 2 percent of modern CPU resources per instance. This strict performance budget necessitates the deployment of O(1) algorithmic complexities per sample, highly optimized polynomial interpolation, and the avoidance of deep iterative solvers in favor of explicit analytical approximations.

This comprehensive report details the optimal DSP architectural roadmap to achieve these goals. It evaluates advanced bandlimited synthesis techniques (PolyBLEP and PolyBLAMP), formulates topology-preserving zero-delay feedback (ZDF) filters with non-linear saturation, establishes the algorithmic topology for a low-overhead effects chain, and defines a parameter-locking automation matrix specifically tailored for tracker interfaces.

The Synthesis Core: Alias-Free Geometric Waveform Generation

The foundation of any chiptune synthesizer relies on geometric waveforms, primarily square, pulse, sawtooth, and triangle waves. However, the naive digital generation of these waveforms (for example, generating a square wave by sampling a continuous function defined as +1 for the first half of a cycle and -1 for the second) produces an infinite series of odd harmonics. According to the Nyquist-Shannon sampling theorem, when these harmonics exceed the Nyquist frequency ($f_s/2$), they reflect back into the audible spectrum as non-harmonic, mathematically unrelated frequencies. This phenomenon, known as foldover or aliasing, destroys the musicality of the oscillator, replacing the aggressive "bite" of analog-style waveforms with harsh digital noise.

To eliminate this aliasing while maintaining the strict computational constraints of the

synthesizer, the architecture must avoid heavy oversampling or the generation of massive bandlimited wavetables, relying instead on localized polynomial correction methods.

Bandlimited Pulse Wave and Dynamic Pulse Width Modulation (PolyBLEP)

The Polynomial Band-Limited Step (PolyBLEP) algorithm is the most computationally efficient method for rendering alias-free discontinuities, such as those found in square and sawtooth waves. Rather than processing the entire waveform through a steep low-pass filter, PolyBLEP operates by generating a naive waveform and then adding a pre-calculated polynomial residual precisely at the points of discontinuity.

The theoretical foundation of PolyBLEP is rooted in the integration of a band-limited impulse train (BLIT). The ideal band-limited step function is equivalent to the sine integral. Integrated polynomial interpolation methods, such as those derived from B-spline basis functions, can successfully approximate this ideal step function. The difference between the naive step and the bandlimited step forms the residual function.

For a pulse wave featuring variable Pulse Width Modulation (PWM), the waveform contains two discontinuities per cycle: a rising edge at phase 0 and a falling edge at a phase equal to the duty cycle w, where 0 < w < 1. The PolyBLEP residual function, evaluated over a normalized phase t and a fractional phase increment dt = f_c / f_s, is defined as a second-order polynomial. The polynomial evaluates to non-zero only within a one-sample radius of the discontinuity.

$$P(t, dt) = \begin{cases} \frac{t}{dt} + \frac{t}{dt} - \left(\frac{t}{dt}\right)^2 - 1.0 & \text{for } 0 \le t < dt \\ \left(\frac{t-1}{dt}\right)^2 + \frac{t-1}{dt} + \frac{t-1}{dt} + 1.0 & \text{for } 1 - dt < t < 1 \\ 0 & \text{otherwise} \end{cases}$$

The synthesis of a bandlimited PWM pulse wave is thus computed by evaluating the naive pulse and injecting the PolyBLEP residual at the rising edge, while subtracting the residual at the falling edge. Because PolyBLEP localizes corrections only to the samples immediately adjacent to the discontinuity, it achieves near-ideal anti-aliasing while requiring only trivial arithmetic operations. This is vastly superior to the Differentiated Parabolic Waveform (DPW) method, which requires large frequency-dependent scaling factors that suffer from precision loss and catastrophic cancellation at low fundamental frequencies. The pseudo-code for the PolyBLEP oscillator implementation is as follows:

```
// Initialization Parameters phase = 0.0 phase_inc = frequency / sample_rate duty_cycle = 0.5 // Dynamically modulatable parameter
// Per-Sample Audio Loop function process_oscillator(): // Advance normalized phase accumulator phase = phase + phase_inc if phase >= 1.0: phase = phase - 1.0
// Generate Naive Pulse (+1.0 or -1.0) naive_out = (phase < duty_cycle) ? 1.0 : -1.0
// Compute PolyBLEP residual for the rising edge at phase 0.0 blep_rise = poly_blep(phase, phase_inc)
// Compute PolyBLEP residual for the falling edge at phase == duty_cycle phase_shifted = (phase - duty_cycle) if phase_shifted < 0.0: phase_shifted = phase_shifted + 1.0 blep_fall = poly_blep(phase_shifted, phase_inc)
// Apply residual corrections to the naive waveform return naive_out + blep_rise - blep_fall function poly_blep(t, dt): if t < dt: t = t / dt return t + t - (t * t) - 1.0 else if t > (1.0 - dt): t = (t - 1.0) / dt return (t * t) + t + t + 1.0 else: return 0.0
```

Triangle and Stepped Triangle Generation (PolyBLAMP)

While PolyBLEP effectively addresses waveforms with discontinuities in amplitude (step functions), a triangle wave possesses continuous amplitude but discontinuous first derivatives (sharp corners). Applying a step correction to a triangle wave yields incorrect harmonic results because the frequency domain characteristics of a slope discontinuity decay at a rate of 1/k^2 rather than 1/k.

To properly anti-alias a triangle wave, the Band-Limited Ramp (BLAMP) or PolyBLAMP method must be utilized. PolyBLAMP is derived by integrating the PolyBLEP residual, effectively smoothing the sharp corners of the triangle wave. The residual is a localized polynomial approximation of the ideal bandlimited ramp, often expressed as a fifth-order polynomial derived from a third-order B-spline basis function. When properly scaled by the slope delta and applied at the phase turning points, PolyBLAMP completely suppresses the upper-harmonic aliasing that occurs when the triangle wave's fundamental frequency approaches the Nyquist limit. An alternative, highly efficient method for generating a bandlimited triangle wave involves synthesizing a PolyBLEP square wave and passing it through a leaky digital integrator ($y[n] = x[n] + R \cdot y[n-1]$), where the coefficient R is close to 1.0. This is followed by a DC-blocking filter to recenter the waveform around zero. Furthermore, classic hardware such as the NES VRC7 and Nintendo's base triangle channel utilize a quantized "stepped" triangle wave (typically 16 or 32 steps). Applying a naive bit-crushing or decimation operation to a clean triangle wave immediately reintroduces aliasing across the entire spectrum. To authentically recreate the stepped triangle without aliasing, the algorithm treats each amplitude quantization step as an individual discontinuity. A localized PolyBLEP algorithm is scaled and injected at each staircase interval, resulting in a harmonically rich, buzzing low-end characteristic of the NES hardware, entirely free of inharmonic fold-over artifacts.

Linear Feedback Shift Register (LFSR) Noise Generation

To achieve authentic 8-bit percussion, snares, and sound effects, the architecture replaces standard continuous white noise generation with a Linear Feedback Shift Register (LFSR). The NES APU utilizes a 15-bit shift register with specific feedback taps to generate its noise channel.

The state of the LFSR is updated either at the global audio-rate clock or a divided sub-clock to create lower sample-rate textures. For a 15-bit sequence generating pseudo-white noise, the shift register performs rapid bitwise operations. The feedback bit is determined by performing an XOR operation on specific taps (e.g., bit 0 and bit 1), and the register is shifted right, with the feedback bit inserted at the most significant bit position.

By altering the tap configuration—for instance, tapping bit 6 instead of bit 1—the sequence length drops drastically from 32,767 steps to a mere 93 steps. This short, repeating sequence generates the distinctly metallic, pitched buzz utilized in retro video game hi-hats, crash cymbals, and engine sound effects.

Wavetable, SoundFont, and Frequency Modulation (OPL) Paradigms

To extend the timbral palette beyond subtractive geometric synthesis, the engine incorporates a low-overhead wavetable loader and a Frequency Modulation (FM) architecture modeled directly

after the Yamaha OPL2/OPL3 (YM3812/YMF262) sound chips.

Lightweight Wavetable and SoundFont Loading

The wavetable module relies on fractional phase accumulators to map the current oscillator phase to a specific index within a loaded single-cycle waveform array. Because the phase accumulator is represented by a floating-point number and the array indices are integers, the engine must interpolate between adjacent samples to avoid quantization noise. While 4-point cubic Hermite interpolation offers superior audio fidelity, linear interpolation is prioritized for raw wavetable lookup to strictly maintain the 1-2% CPU budget constraint across high voice counts. To handle arbitrary wavetables and SoundFont formats without introducing aliasing, the architecture employs offline pre-processing. Upon loading a wavetable, the engine generates a mipmap of bandlimited tables. By executing an offline Fast Fourier Transform (FFT), removing harmonics above the Nyquist frequency for different target pitch ranges, and executing an Inverse Fast Fourier Transform (IFFT), the engine creates a set of pre-filtered wavetables. During real-time playback, the oscillator dynamically crossfades between the appropriate tables based on the current fundamental frequency, ensuring completely alias-free arbitrary waveform playback at a minimal real-time CPU cost.

Phase Modulation and OPL-Style Synthesis

True to the Yamaha hardware, the FM synthesis section is mathematically implemented as Phase Modulation (PM). The output of a modulator operator dictates the phase offset of the carrier operator before the carrier's waveform is evaluated. The fundamental equation for a two-operator phase modulation pair is y(t) = \sin(\omega_c t + I \cdot \sin(\omega_m t)), where I represents the modulation index governing spectral bandwidth.
To maintain ultra-low overhead, the system emulates the mathematical shortcuts of the original Yamaha OPL chips, which avoided expensive floating-point multiplication by utilizing logarithm and exponential lookup tables (LUTs). Furthermore, the base waveform is derived from a high-resolution 256-entry lookup table representing only a *quarter* of a sine wave. The remaining quadrants are extrapolated on the fly via mathematical symmetry, heavily conserving cache memory. The architecture replicates the OPL2's selectable operator waveforms by applying simple logical rules to the phase accumulator prior to table lookup:

- **Unmodified Sine:** Standard full-wave readout.
- **Half Sine:** Negative phase outputs are clamped to zero. * **Absolute Sine:** Negative phase outputs are inverted, creating a full-wave rectified signal. * **Quarter Sine (Pseudo-Sawtooth):** Odd quadrants are muted entirely, creating alternating positive pulses. These 2-to-4 operator matrices allow for a wide range of algorithms, routing modulators into carriers in series, in parallel, or utilizing self-modulating feedback loops to generate noise and chaotic textures.

The Programmable Effects Pipeline

The post-oscillator signal flow routes through a modular effects chain specifically designed for deep tracker automation. The architecture requires a filter, delay line, and reverberation unit capable of handling rapid, discontinuous parameter modulation (such as per-step parameter

locks) without exhibiting mathematical explosion, structural instability, or zipper noise.

Linear Trapezoidal Zero-Delay Feedback (ZDF) State-Variable Filter

Traditional digital biquad filters (e.g., Direct Form I or II) exhibit severe numerical instability when their cutoff frequencies are modulated at high audio rates. Additionally, older implementations of the digital State-Variable Filter (SVF), popularized by Hal Chamberlin, suffer from high-frequency warping and become completely unstable when the cutoff frequency approaches one-sixth of the sampling rate (f_s/6). To resolve this and ensure the filter can withstand aggressive automation from LFOs and step sequencers, the architecture implements the Linear Trapezoidal Integrated State Variable Filter, pioneered by Andrew Simper of Cytomic. This topology-preserving transform (TPT) filter resolves the implicit delay-free loop of an analog SVF circuit by replacing naive integration with trapezoidal integration. The resulting algorithm allows fully independent control over the cutoff frequency and resonance (Q), produces simultaneous low-pass, high-pass, band-pass, and notch outputs, and remains unconditionally stable under rapid modulation up to the Nyquist limit. The pre-warped coefficients for the Simper SVF are computed analytically:
g = \tan\left(\pi \frac{f_c}{f_s}\right) k = \frac{1}{Q} a_1 = \frac{1}{1 + g(g + k)} a_2 = g \cdot a_1 a_3 = g \cdot a_2
To replicate the aggressive, self-oscillating "screaming" character of analog MS-20 or Moog ladder filters, non-linear saturation must be embedded directly into the filter's feedback loop. By passing the internal integrators or the resonance feedback signal through a hyperbolic tangent function (tanh), the filter's resonance peaks are dynamically compressed. This prevents infinite amplitude explosion when self-oscillating, while simultaneously generating rich, odd-order harmonic distortion.
To strictly maintain the CPU budget, calling the standard library trigonometric tanh() function on a per-sample basis is prohibited. Instead, the architecture utilizes a fast analytical approximation (such as a Padé approximant or a rational polynomial wave-shaper) to mimic the clipping curve. The pseudo-code for the non-linear SVF audio-rate processing loop is as follows:
// Process Filter called per audio sample function process_filter(input, drive, filter_mode): // Apply soft saturation to the input signal to emulate analog overdrive input = fast_tanh_approximation(input * drive)
// Calculate intermediate variables v3 = input - ic2eq v1 = a1 * ic1eq + a2 * v3 // Band-pass output v2 = ic2eq + a2 * ic1eq + a3 * v3 // Low-pass output
// Update the trapezoidal integration internal states ic1eq = 2.0 * v1 - ic1eq ic2eq = 2.0 * v2 - ic2eq
// Return the selected filter topology if filter_mode == LOW_PASS: return v2 if filter_mode == HIGH_PASS: return input - (k * v1) - v2 if filter_mode == BAND_PASS: return v1 if filter_mode == NOTCH: return input - (k * v1)

Delay Lines and Cubic Hermite Interpolation

The digital delay line forms the architectural basis for echo, chorus, and flanger effects. Modulating the read pointer of a delay buffer to create these pitch-shifting effects—or abruptly changing the delay time via step-sequencer automation—requires precise sub-sample fractional interpolation.
Linear interpolation, while computationally cheap, acts as an unintentional low-pass filter, severely dampening high-frequency content when the fractional delay offset rests near 0.5. To

maintain pristine audio fidelity and prevent digital clicking during arbitrary millisecond offsets, the engine utilizes a 4-point Cubic Hermite Spline interpolation algorithm.
Given a fractional offset t \in [0, 1) and four adjacent audio samples y[n-1], y[n], y[n+1], y[n+2] retrieved from the circular ring buffer, the Hermite interpolator ensures C^1 mathematical continuity, meaning both the amplitude and the first derivative of the waveform transition smoothly across the sample boundaries. The computationally optimized basis functions are calculated as:
a = -0.5 y[n-1] + 1.5 y[n] - 1.5 y[n+1] + 0.5 y[n+2] b = y[n-1] - 2.5 y[n] + 2 y[n+1] - 0.5 y[n+2] c = -0.5 y[n-1] + 0.5 y[n+1] d = y[n] \text{Interpolated Output} = a t^3 + b t^2 + c t + d
This method provides an optimal structural balance between the prohibitive computational cost of 8-point Sinc interpolation and the destructive, dulling artifacts of linear interpolation.
Furthermore, when grid-synced delay divisions (e.g., 1/8th dotted, 1/16th) are automated via parameter locks, a secondary one-pole smoothing algorithm glides the absolute read-pointer position over a defined millisecond window. This intentional smoothing allows for distinct tape-delay-style Doppler pitch-shifting effects rather than abrupt, glitchy jumps.

Algorithmic Reverberation: Schroeder-Moorer Topology

Modern convolution reverbs or highly dense Feedback Delay Networks (FDN) require deep matrix multiplications that rapidly exceed the stringent 1-2% CPU budget. Furthermore, they often sound too realistic and pristine for a chiptune aesthetic. Therefore, the architecture employs a highly optimized algorithmic topology derived from the early digital reverberators designed by Manfred Schroeder and James Moorer. The Schroeder-Moorer topology utilizes a parallel bank of Feedback Comb Filters (FBCF) routed into a series cascade of All-Pass Filters (APF). The parallel comb filters simulate the discrete early reflections and the geometric modes of a physical room. However, comb filters inherently possess a metallic, ringing frequency response. To diffuse this ringing, the signal is passed through the series all-pass filters, which function as "impulse diffusers." These all-pass sections rapidly multiply the echo density, smearing the transients and creating a smooth reverberant tail without heavily altering the overall frequency amplitude response. A standard configuration optimized for retro character and minimal CPU footprint includes:

Reverb Component	Filter Count	Delay Time Range	Target Function
Comb Filters (FBCF)	4 to 6 Parallel	30ms – 80ms	Simulates early reflections; feedback coefficients scaled to define global RT_{60} decay time.
All-Pass Diffusers (APF)	2 to 3 Series	1ms – 6ms	Smears transients to increase echo density; fixed gain coefficient typically set at g = 0.7.

To prevent undesirable resonance buildup or metallic "flutter," the delay times (measured in precise samples) of both the comb and all-pass filters must be mutually prime numbers, ensuring that specific phase alignments do not constructively sum at regular intervals.
Modulating the comb filter delay lengths with a slow, low-depth LFO provides a synthetic chorus to the reverb tail, further smoothing the response and simulating the natural acoustic air

fluctuations of large spaces.

Tracker Automation and the Modulation Matrix

In a tracker-based workflow, sequencer steps routinely trigger sudden parameter jumps—referred to as parameter locking or "p-locking." Instantly snapping a filter cutoff from 10 kHz to 200 Hz, or leaping from one delay time to another at the boundary of a sequencer step, generates broadband impulses that manifest as digital clicks or "zipper noise." To maintain artifact-free playback, the automation handling and modulation matrix must decouple control signals from the audio pipeline.

Per-Step Parameter Locks and Boundary Smoothing

To handle instantaneous step boundaries gracefully, all automated parameters undergo control-rate smoothing via a first-order low-pass filter (leaky integrator). When a parameter lock requests a new value x_{target}, the internal control variable y_{ctrl} approaches the target asymptotically:
y_{ctrl}[n] = y_{ctrl}[n-1] + \alpha (x_{target} - y_{ctrl}[n-1])
The smoothing coefficient \alpha dictates the slew rate. For per-step parameter locks, a fixed time constant of roughly 5 to 15 milliseconds is applied. This duration is mathematically fast enough to preserve the rhythmic precision expected from a hard-quantized step sequencer, but long enough to roll off the high-frequency energy of the discontinuity, ensuring the transition occurs entirely within the sub-audio spectrum.

Audio-Rate vs. Block-Rate Processing

To satisfy CPU restrictions, the modulation matrix employs a dual-rate processing structure. Audio-Rate (per-sample) processing is reserved strictly for fast transient modulations—such as audio-rate FM, rapid PWM, and hard-sync oscillator pitch tracking. These parameters are evaluated identically across every frame of the audio buffer.
Conversely, Control-Rate (per-block) processing is used for Macro LFOs, overall filter cutoff sweeps, and global effects mix parameters. These variables are computed only once per block (e.g., every 32 or 64 samples). The final control values are then linearly interpolated across the block during the audio rendering phase. This guarantees smooth parameter transitions while avoiding the need to evaluate expensive exponential frequency curves or logarithm calculations on a per-sample basis. Furthermore, LFOs maintain dedicated phase accumulators capable of operating in dual timing modes. In *Host-Synced* mode, the phase is mathematically locked to the DAW or tracker transport, evaluating frequency relative to the BPM grid. In *Free-Run* mode, the phase is derived from absolute elapsed milliseconds, ensuring modulation independence from temporal host fluctuations.

DSP Signal Flow and Parameter Mapping

The modular structure dictates the signal flow sequentially, ensuring predictable headroom and staging: MIDI / Tracker Input -> Voice Allocation Matrix -> Oscillators (PolyBLEP / FM / LFSR / Wavetable) -> Voice Envelopes (ADSR) -> Polyphonic Mix Bus -> State-Variable Filter (SVF) -> Delay Line -> Reverb -> Master Output Clip.

To ensure consistent logic across the tracker step-sequencer and external VST parameter interfaces, values are mapped using standardized mathematical ranges, scaling curves, and smoothing directives.

Module	Parameter	Value Range	Scaling Curve	Smoothing Directive	Sync Modes
Oscillator	Pitch	10.0 – 12,000 Hz	Exponential	Variable Glide	Free / MIDI Note
Oscillator	PWM Duty	1\% – 99\%	Linear	5 ms LPF	Free / Audio-Rate
Filter	Cutoff (f_c)	20 – 20,000 Hz	Logarithmic	10 ms LPF	Block-Rate Interpolated
Filter	Resonance (Q)	0.707 – 25.0	Exponential	10 ms LPF	Block-Rate
Filter	Drive	1.0 – 10.0	Linear	15 ms LPF	Block-Rate
Delay	Time	1 – 2,000 ms	Linear	50 ms Doppler Slew	Milliseconds / BPM Grid
Reverb	Decay (RT_{60})	0.1 – 10.0 sec	Exponential	20 ms LPF	N/A
LFO	Rate	0.1 – 100 Hz	Logarithmic	None (Direct)	Milliseconds / BPM Grid

Performance Benchmarks and Hardware-Level Optimization

Achieving elite high-fidelity DSP within a rigid 1-2% CPU threshold mandates strict lower-level software engineering considerations, prioritizing instruction cache coherence and mathematical simplification.

1. **Trigonometric Avoidance:** Standard math library functions such as sin(), tan(), and pow() require massive clock-cycle overhead when placed inside per-sample audio loops. For the OPL FM algorithms, pre-computed fractional lookup tables entirely substitute real-time trigonometric math. For the SVF, the tangent calculation required for the cutoff frequency coefficient g is strictly isolated to the block-rate parameter update function, evaluating only when a modulation source actively changes the cutoff value.
2. **Explicit Saturation Approximations:** The tanh() non-linearity utilized for filter drive is approximated using rational polynomials or Padé approximants (e.g., x \cdot (27 + x^2) / (27 + 9x^2)). This mathematical shortcut perfectly preserves the soft-clipping dynamic range and odd-harmonic saturation curve while completely bypassing the costly standard library execution.
3. **Memory Access & Cache Coherence:** The Schroeder-Moorer algorithmic reverb and Hermite fractional delays require multiple overlapping read and write pointers across large delay line buffers. By allocating these circular buffers contiguously in memory and utilizing bitwise masking on buffer lengths precisely sized to powers of two (e.g., using index & (2048 - 1) instead of the modulo operator % 2048), pipeline stalling is prevented and critical L1/L2 cache misses are eradicated.
4. **SIMD Vectorization:** Where applicable—such as executing multiple voices of oscillator polyphony or processing the left and right channels of the stereo delay lines concurrently—operations are written to leverage AVX or NEON Single Instruction, Multiple Data (SIMD) vector instructions. This processes floating-point arrays in parallel, effectively

cutting the active processing time of the effects chain in half and easily maintaining the sub-2% CPU threshold per plugin instance.

Conclusions

The DSP architecture outlined for this 8-bit synthesizer and step-automated effects module successfully bridges the gap between mathematically authentic vintage hardware emulation and robust, modern digital performance standards. By integrating localized PolyBLEP and PolyBLAMP polynomial correction algorithms, the system synthesizes geometric waveforms entirely free from the destructive aliasing of naive digital generation. Expanding this tonal footprint through LFSR noise arrays and OPL-inspired quarter-sine phase modulation effectively encapsulates the highly sought-after artifacts of the golden era of chiptune sound design. Simultaneously, the programmable effects chain ensures total audio stability under the extreme parameter-lock sequencing demanded by modern tracker workflows. The incorporation of Andrew Simper's Trapezoidal State-Variable Filter guarantees artifact-free, audio-rate cutoff modulation without risk of structural explosion, while the deployment of 4-point Cubic Hermite Spline interpolation enables pristine, click-free delay line manipulations. By anchoring these algorithms in rigid, constant-time O(1) complexity processing, optimizing transcendental functions through analytical approximations, and decoupling control signals via block-rate smoothing, the engine successfully achieves elite audio fidelity well within the strictest CPU performance boundaries.

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